

FM321 Project

A Comparative Analysis of Commodity Markets and the Platform-based Steam Marketplace

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Abstract

Commodity markets are organised exchanges for trading largely fungible goods, and have been extensively studied due to their distinctive volatility dynamics and links to equity markets. This raises the question of whether other, non traditional markets with similar structural features exhibit comparable price and risk behaviour. This paper examines this question by analysing the Steam Community Market, a platform mediated digital marketplace introduced by Valve in 2012, where users trade in game items whose value is primarily derived from scarcity rather than physical delivery or legal ownership. Using price data from the CS2 skin market, this study compares the stylised facts, volatility dynamics, and risk characteristics of the Steam marketplace with those observed in traditional commodity markets. We evaluate the performance of standard volatility forecasting models, including GARCH type specifications, with a particular focus on APARCH and FIGARCH models, and assess the effectiveness of Value at Risk as a risk measure using both historical simulation and parametric approaches. The models are further evaluated through formal backtesting using coverage tests. The results provide evidence of volatility clustering, heavy tails, and long memory behaviour in Steam market returns, mirroring key empirical features commonly documented in commodity markets. Volatility models that are effective in commodity market analysis are shown to perform well in the Steam marketplace, and the resulting Value at Risk estimates are broadly consistent with findings in the existing literature. Institutional constraints such as transaction fees and restricted fund redemption prevent direct comparability in terms of investment feasibility, nevertheless the similarities in price dynamics and risk behaviour suggest a meaningful connection between the two markets.

1 Introduction

Commodity markets are organised trading venues for standardised and largely fungible goods. They have been a major focus of modern research both due to their interconnectedness with equity markets [9] and their role in hedging risk. This raises the question of whether other markets that share similar structural characteristics also exhibit comparable price dynamics. One such market, the CS2 skin marketplace, is the focus of this analysis. Steam was created in 2003 as a digital content distribution platform. In 2012, Valve, the company behind Steam, introduced a marketplace that allows users to buy and sell in-game items. Among the largest segments of this marketplace is the CS2 skin market, where users trade cosmetic in-game items. This market has grown rapidly, reaching a peak valuation of approximately \$6 billion.¹ Assets traded on the Steam marketplace are cosmetic digital items whose value is primarily derived from their scarcity. These items are typically acquired through a lottery-based mechanism, in which the most common drops occur with a probability² of approximately 80%, while the rarest items are obtained with a probability of about 0.26%.

This analysis proceeds in two parts. First, it examines the CS2 skin marketplace to identify patterns in prices and volatility. Second, it evaluates the extent to which these dynamics resemble those observed in traditional commodity markets. In doing so, the paper explores whether quantitative risk-modelling

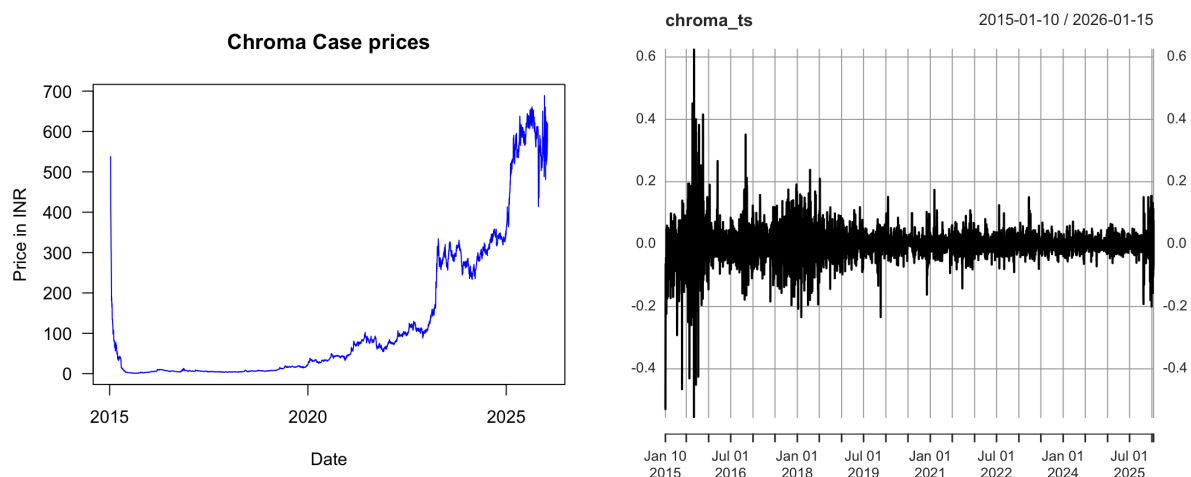
¹Counter-Strike skins market soars to new peak of \$5.78 billion, <https://esports-news.co.uk/2025/10/08/counter-strike-skins-market-soars-to-new-peak-of-5-78-billion>

²<https://siege.gg/news/behind-the-drop-rates-how-cs2-case-odds-really-work>

techniques commonly applied in commodity markets can be meaningfully extended to platform-mediated online marketplaces such as Steam. This work represents the first empirical analysis of price dynamics in the Steam Community Market and therefore serves as a foundation for future research into such emerging markets. Existing research has examined volatility behaviour in related markets, including commodity and NFT markets [8]; this study contributes to that literature by extending the analysis to a large-scale, platform-based digital marketplace. The analysis begins with the selection of a particular asset and a discussion of the data sources and data quality. We then document the key stylised facts and basic empirical properties of the assets. The analysis then evaluates the effectiveness of volatility forecasting models GARCH, APARCH, and FIGARCH. The analysis subsequently considers risk measurement through Value at Risk (VaR), employing both historical simulation methods and parametric approaches based on GARCH models. Finally, we focus on comparing observed trends in commodity markets with those in the Steam marketplace.

2 Preliminary Analysis

Users can trade cosmetic items (“skins”) on the Steam Marketplace by posting sell offers, which specify the price at which they are willing to sell an item, and buy orders, which indicate the price buyers are willing to pay. The assets analysed in this study are known as cases. Cases function as a lottery mechanism through which players can obtain rare skins. Players acquire cases randomly as in-game drops after a certain amount of playtime. However, opening a case requires purchasing a key, which typically costs significantly more than the market value of the case itself. As a result, it is in the interest of users to sell cases on the marketplace rather than open them. Following this, the case selected for analysis is the Chroma Case, released in 2015. This case was chosen due to its ten-year trading history and consistently high trading volume. The data on the price of the Chroma Case was obtained directly from the Steam site³; it lists the price at which items are sold and the volume sold. I aggregated and sorted this data to prepare it for analysis. I started by plotting the price history and log returns.



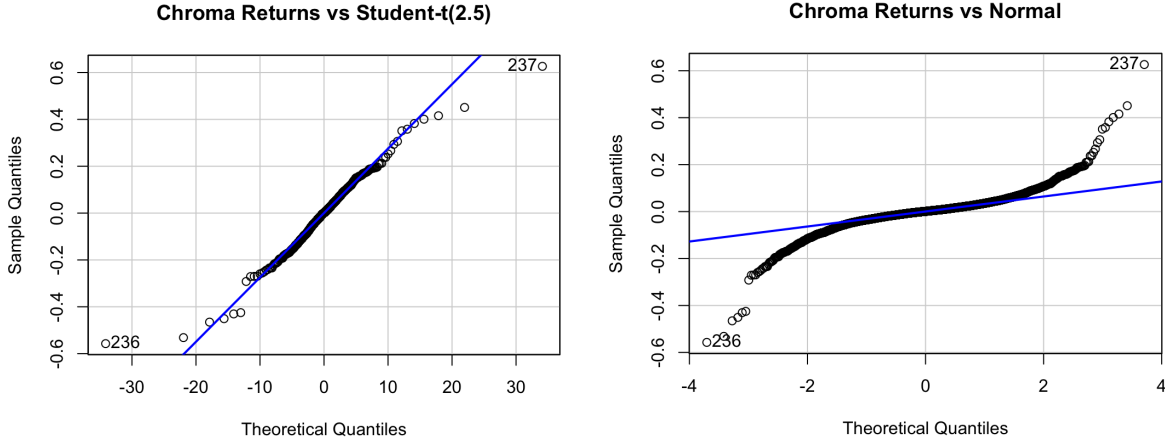
The Chroma Case, released in 2015, was initially scarce. This temporary scarcity led to a high initial market price, which quickly declined as supply increased and the case reached a more stable trading level. The volatility time series shows clear evidence of volatility clustering. However, this behaviour is unlikely to be explained by mechanisms such as the leverage effect, since concepts like debt-to-equity ratios are not applicable to in-game assets. Instead, volatility clustering is more plausibly explained by shifts in user expectations. The descriptive statistics broadly resemble those observed in equity markets; however, the standard deviation is notably higher, reflecting the more volatile nature of returns in the case market. A kurtosis value greater than 3 suggests that the returns do not follow a normal distribution.

³<https://steamcommunity.com/market/listings/730/Chroma%20Case>

Statistic	Value
Mean	−.000098
Standard Error	0.05476
Minimum	−0.55721
Maximum	.626455
Skewness	−0.24941
Kurtosis	17.24563

Table 1: Log-returns descriptive statistics

This observation is reinforced by the QQ-plots and the Jarque–Bera test. The Jarque–Bera test yields a test statistic of 5876 for the return data, and at the 1% significance level the critical value is 9.21034; therefore, we reject the null hypothesis of normality, as the test statistic is much larger than the critical value. Furthermore, the QQ-plots indicate non-normality and suggest that a t -distribution with 2.5 degrees of freedom is more suitable for modelling the distribution of returns.



3 Volatility Forecasting

To transition into the comparative analysis of commodity markets and the Steam Marketplace, it is beneficial to consider how models commonly used in commodity market analysis perform when applied to volatility forecasting in Steam markets. In this context, analysis by Alfeus and Nikitopoulos [1] indicates that a FIGARCH (Fractional Integrated GARCH) model is successful in detecting the long-memory characteristics of commodity markets and performs marginally better over other models in long time horizons. Further, as shown by Giot and Laurent [4], when analysing market risk using a Value-at-Risk (VaR) approach, a APARCH model performs best. To begin, we examine how a GARCH(1,1) model performs and then observe whether FIGARCH and APARCH provide improvements. We use the package `ugarchspec()` and `ugarchfit()` to obtain the relevant forecasting models. It is relevant to note that we assume normality, rather than a t distribution, due both to computational considerations and the small window size.

3.1 FIGARCH

To understand what the FIGARCH model does, we must first take note of the fact that a GARCH(p, q) process can be written as (3), which is done through the usage of a backshift operator $Lx_t = x_{t-1}$, with x_t being returns

$$x_t = \varepsilon_t \sigma_t \quad (1)$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i x_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad (2)$$

$$= \alpha_0 + \alpha(L)x_t^2 + \beta(L)\sigma_t^2 \quad (3)$$

Now this means that we can write our GARCH process as an ARMA(m, p) process with $m = \max\{p, q\}$ and $y_t = x_t^2 - \sigma_t^2$

$$[1 - \alpha(L) - \beta(L)](1 - L)x_t^2 = \alpha_0 + [1 - \beta(L)]y_t \quad (4)$$

Using the approach as done by T. V. Ramanathan [7], the fractionally integrated GARCH(p, q), or FIGARCH(d, p, q), class of models is obtained by replacing the first difference operator $(1 - L)$ in (4) with the fractional differencing operator $(1 - L)^d$, where d is a fraction such that $0 < d < 1$. Thus, the FIGARCH class of models can be obtained by considering

$$[1 - \alpha(L) - \beta(L)](1 - L)^d x_t^2 = \alpha_0 + [1 - \beta(L)]y_t. \quad (5)$$

3.2 Model performance

We would expect APARCH and FIGARCH to both be statistically significant enough to be preferred over a GARCH(1,1). To confirm this, we estimate the parameters for each of the models.

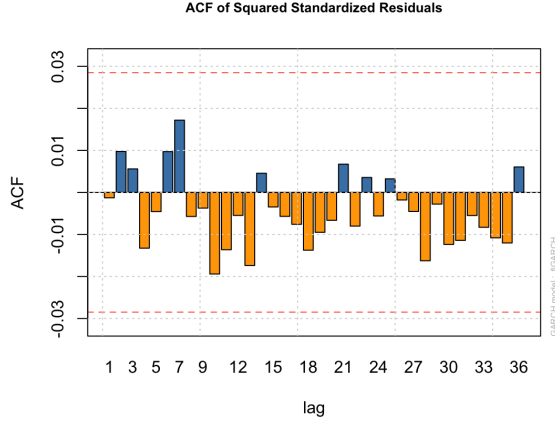
Parameter	GARCH(1,1)	APARCH(1,1)	FIGARCH(1,1)
ω	0.000014500	0.0000020057	0.0000070026
α_1	0.07396691	0.04878647	0.19113820
β_1	0.92138800	0.92957560	0.94531440
γ_1	—	0.10205980	—
δ	—	2.5544030	—
d	—	—	0.99999880

As a visual inspection, we can see that the additional parameters in both APARCH and FIGARCH appear to differ sufficiently from those of the original model to be statistically significant. To formally validate this, we use the likelihood ratio test.

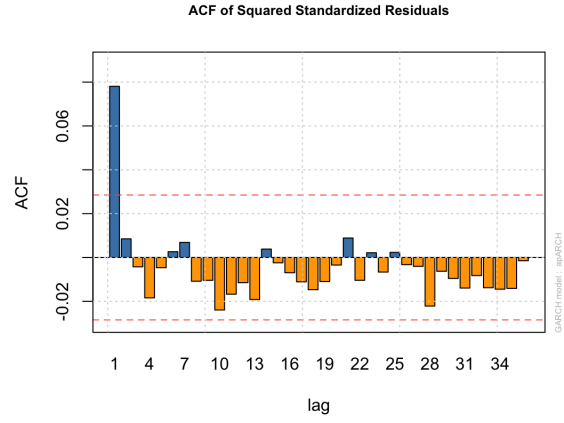
$$LR = 2(\log \mathcal{L}_U - \log \mathcal{L}_R) \sim \chi_{(\text{number of restrictions})}^2$$

For the FIGARCH model, we obtain a likelihood ratio statistic of $LR = 532.9483$, which is compared against the 1% critical value $\chi_{1,1\%}^2 = 6.634897$. Since the test statistic exceeds the critical value, we reject the null hypothesis and conclude that the FIGARCH model provides a statistically significant improvement over the GARCH model. Similarly, for the APARCH model, we obtain $LR = 21.79304$, which exceeds the 1% critical value $\chi_{2,1\%}^2 = 9.21034$, again indicating a statistically meaningful improvement. These results support the view that volatility in the returns of the Chroma Case exhibits behaviour analogous to volatility patterns commonly observed in commodity markets [1].

When analysing the APARCH and FIGARCH models, we would expect no autocorrelation in the residuals. In the plots below, we observe that FIGARCH performs effectively, with all bars of the FIGARCH residual ACF falling within the confidence bands. It appears to perform better than APARCH, as most ACF values of FIGARCH at different lags are lower than those of APARCH. While the absence of residual autocorrelation provides supportive evidence for the adequacy of the models and reinforces the narrative of similarity between this market and commodity markets, these findings also motivate the use of FIGARCH and APARCH models in subsequent Value-at-Risk analysis. This section has demonstrated not only which models are most effective for forecasting volatility in the Steam Marketplace, but also that the existing literature on commodity markets provides a strong starting point for analysing non-conventional markets. As financial activity expands into areas such as cryptocurrencies and non-fungible tokens [3], this section demonstrates the applicability of commodity market focused research to emerging markets.



FIGARCH(1,1)



APARCH(1,1)

4 Risk Forecasting

This section examines the effectiveness of Value-at-Risk (VaR) as a risk measure for a portfolio composed entirely of a Chroma Case. Before proceeding with the VaR analysis, it is essential to assess the feasibility of trading and maintaining such a portfolio in the Steam Marketplace, as there exist constraints such as the Steam Wallet, ownership structures, and transaction costs. Following this discussion, both parametric and non-parametric approaches to VaR estimation are applied across different time horizons. Finally, these methods are backtested and their performance is evaluated relative to VaR results commonly observed in commodity markets.

4.1 Investment feasibility

There are two primary issues affecting the viability of investment in the Steam Marketplace. The first concerns the storage and transferability of funds. As Steam operates primarily as a digital distribution platform, transactions are conducted through the Steam Wallet system. While users can deposit funds into their wallets, direct withdrawal is not possible. As a result, Steam Wallet balances are not directly transferable into real money. Although indirect methods exist through third-party platforms, these introduce additional risks, including potential violations of Steam's policies and exposure to fraudulent or unregulated intermediaries. The second issue relates to transaction costs. Selling items on the Steam Marketplace incurs a transaction fee of approximately 15%, which is substantially higher than fees typically observed in conventional financial or commodity markets. This level of transaction cost significantly reduces the practicality of active trading and investment portfolios. Despite large-scale portfolio investment not being practical, the Steam Marketplace provides a valuable case study for examining how community-driven markets can display volatility dynamics similar to those observed in established commodity markets. Consequently, while direct exploitation of Value-at-Risk through portfolio construction may be limited, studying the behaviour of VaR in this context remains analytically insightful.

4.2 Value-at-Risk Analysis

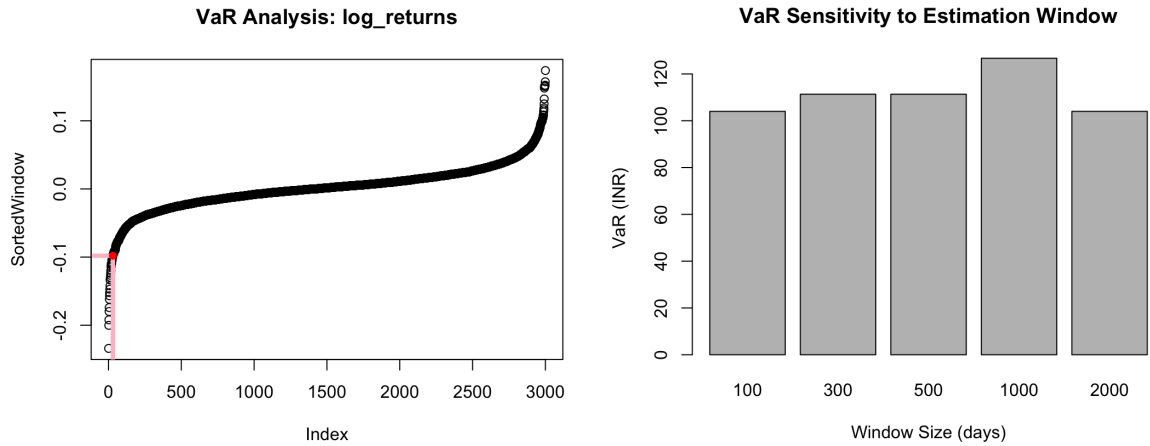
To evaluate the effectiveness of Value at Risk (VaR) as a risk measure, we consider multiple modelling approaches. These include a non-parametric historical simulation method, as well as parametric approaches that require distributional assumptions and parameter estimation. We calculate the VaR with a 1-day holding period.

The historical simulation approach requires the specification of a time window and a probability level; these are used to compute the corresponding VaR estimates. For the parametric approaches, building on Section 3, we assume normality of returns and that volatility follows either a FIGARCH or an APARCH

process. We also note that the formula for VaR based on simple returns is approximately equivalent to that for log returns; as such, we can use the following equation.

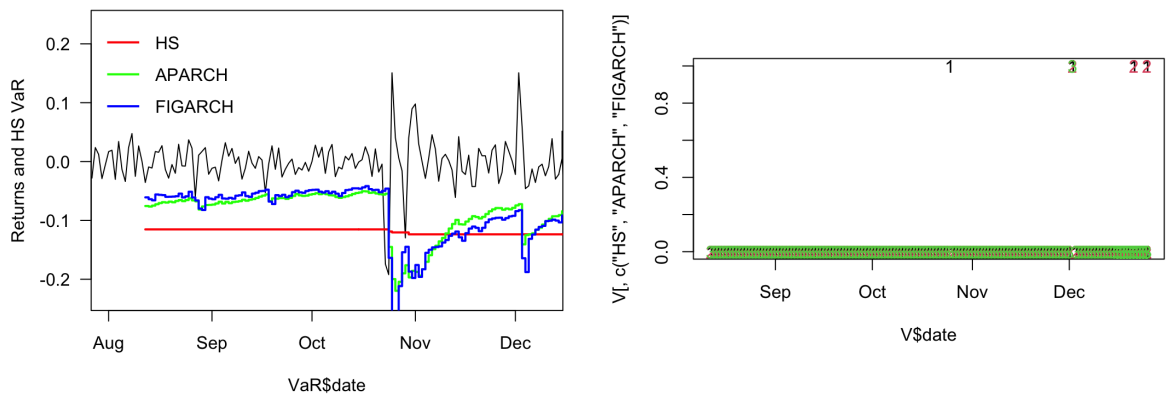
$$\widehat{\text{VaR}}_{t+1} = -\hat{\sigma}_{t+1} \Phi^{-1}(\rho) v,$$

where $\hat{\sigma}_{t+1}$ is the conditional volatility at time $t+1$, Φ^{-1} denotes the inverse standard normal cumulative distribution function, ρ is the probability level, and v denotes the portfolio value.



We calculate Value at Risk using the historical simulation method with a probability level of 1% for a portfolio value equal to 1000 INR. A window of length T equal to 3000 is used, which, following Hendricks [5], should be sufficient to obtain accurate estimates of Value at Risk. Our R programme produces a Value at Risk of 98 INR on a given day. Given the large number of observations, historical simulation provides a reliable benchmark for assessing VaR. The diagram on the left illustrates how the HS method is constructed, while the diagram on the right displays the sensitivity of the HS-based Value at Risk estimates to different window lengths. The Value at Risk appears to shift by approximately 10–15% as the window length changes. This highlights the necessity of using a large sample size in order to minimise the variability of the Value at Risk estimates.

We move on to estimating VaR using a parametric approach based on the APARCH and FIGARCH models, under the assumptions outlined above, with the sole difference that the portfolio value is set to 1 INR. Due to computational complexity, we do not estimate VaR over the full 3000-day sample. Instead, we compute VaR for 300 observations, corresponding to approximately one year of data. We then zoom in on the period 2025-08-01 to 2025-12-10, which is of particular interest due to elevated volatility, and assess the different VaR models. The figure on the left shows this zoomed-in period.



The historical simulation method does not perform well, as it remains relatively flat throughout the period. This is most likely due to the limited sample size of 300 observations, which restricts its ability to adapt to changing market conditions. When comparing the FIGARCH and APARCH models, we observe that FIGARCH tends to underestimate risk during periods of relative stability and overestimate risk during periods of heightened volatility. This behaviour is consistent with the findings of [1], which suggest that FIGARCH requires long time horizons to perform as effectively as alternative volatility models. To assess the models' performance empirically, we employ a backtesting approach based on the coverage test, which examines whether the expected number of violations, defined as observations falling below the estimated Value at Risk, matches the observed number of violations.

The figure above on the right shows the number of violations for each model. However, this alone does not illustrate the feasibility of the models. To address this, we apply the Bernoulli coverage test⁴ at the 5% significance level. This yields test statistics of 0.027, 0.135, and 2.774 for the HS, APARCH, and FIGARCH models, respectively. Comparing these statistics to the critical value $\chi^2_{1,5\%} = 3.841459$, we fail to reject the null hypothesis for all three models.

5 Comparison to Commodity Markets

When examining similarities in terms of risk and forecasting, we observe that models which perform well in capturing volatility dynamics in commodity markets also perform well in the Steam marketplace. Furthermore, when considering Value at Risk as a risk measure, the results obtained are consistent with findings in the existing literature. Taken together, these results suggest that a meaningful link can be established between the functioning of traditional commodity markets and markets in which non fungible assets are traded. Building on this, existing research on commodity markets by Kang [6] and Wang [8] shows that commodity markets exhibit established links with equity markets. This suggests that it is plausible for emerging markets, particularly those experiencing increasing capitalisation, to exhibit similar spillover effects and interactions with equity markets.

6 Conclusion

This paper presents an empirical study of the common characteristics shared by commodity markets and the non-traditional market of the Steam marketplace. Specifically, we compare stylised statistics, the suitability of similar volatility models, and the effectiveness of risk measures across these markets. Using results from the existing literature on commodity markets, we observe features such as volatility clustering, a characteristic that is present in commodity markets as noted by Ding [2]. Furthermore, the results highlight the applicability of volatility models that have proven effective in commodity markets, namely the APARCH [4] and FIGARCH [1] models. Their ability to capture volatility dynamics ultimately leads to reliable Value at Risk estimates. While strong similarities with commodity markets are present, the Steam marketplace operates under institutional constraints, including non-transferable wallet balances and high transaction fees. These features limit investment feasibility and question the applicability of the research. Therefore this study should be seen as a case study of an emerging market, instead of being used as guidance for investment strategies.

By extending commodity market volatility and risk modelling techniques to a platform mediated digital marketplace, this paper contributes to the growing literature on the commercialisation of non traditional assets. The Steam Community Market provides a interesting case study for studying the return dynamics of emerging asset classes, including digital collectibles and non fungible assets. Areas for further research include analysing the price dynamics of portfolios composed of multiple cases, which may be more indicative of broader market trends. In addition, there is scope to study potential arbitrage opportunities within the market, particularly given the absence of formal trading mechanisms.

⁴The null hypothesis is that the model is correctly specified at the chosen probability level p .

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